

Reference: Using Vernier Calipers

This document explains how to use Vernier calipers to measure length.

Example of how to use Vernier calipers

A caliper is a tool used to take precise length measurements. The calipers we will be using typically have a resolution of either 0.1 mm or 0.05 mm (meaning that this is the size of the smallest division). Figure 1 shows a caliper with the important parts labelled.

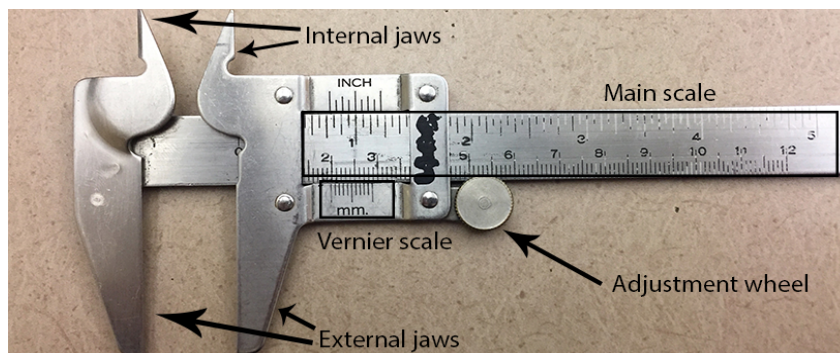


Figure 1: Vernier caliper with relevant parts identified.

To use the caliper, separate the jaws by either pulling them apart or using the adjustment wheel. Place the object you'd like to measure between the external jaws and close them onto the object. (If you are measuring an inside distance, first compress the jaws instead and then separate the calipers until the internal jaws reach the two inside points that you would like to measure the distance between.)

To read the measurement:

- Look at the first tick mark on the Vernier scale. If this tick mark is lined up perfectly with a tick mark on the main scale, this main scale reading will be your measurement.
- If not, look at the first main scale tick mark to the left of the Vernier scale reading. This will be your main reading. For example, in Figure 2, the first tick mark of the Vernier scale is between 2.2 and 2.3, so your main reading is 2.2 cm.
- Follow the Vernier scale until a tick mark lines up perfectly with one on the main scale. This Vernier tick mark will be your second reading, which you should add onto the end of the first reading. Again in Figure 2, the Vernier tick mark representing 6 lines up perfectly with the main scale, so the overall reading is 2.260 ± 0.005 cm.

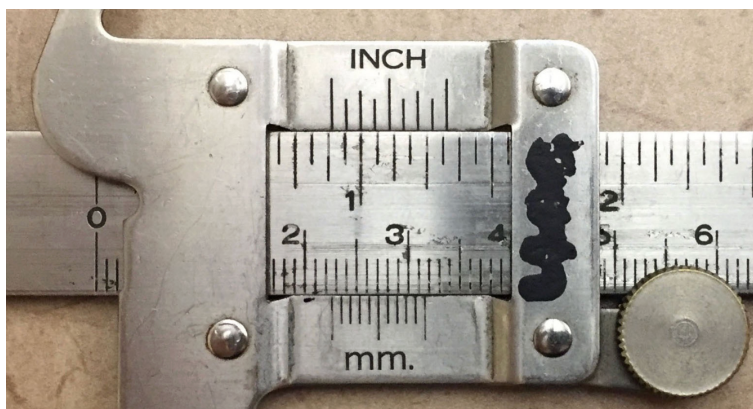


Figure 2: Example measurement with a caliper.

Practice Quiz

Three different measurements are shown in the figures below. What is the reading on each of the calipers in the figures? You can check your answers on the next page.

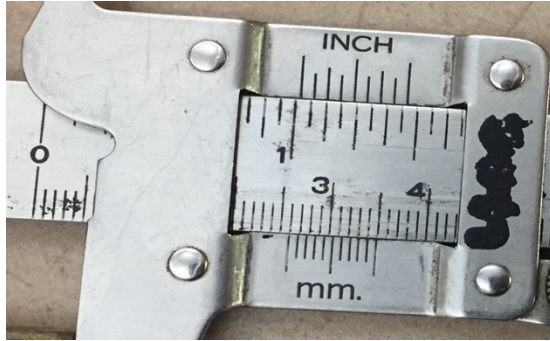


Figure 3



Figure 4

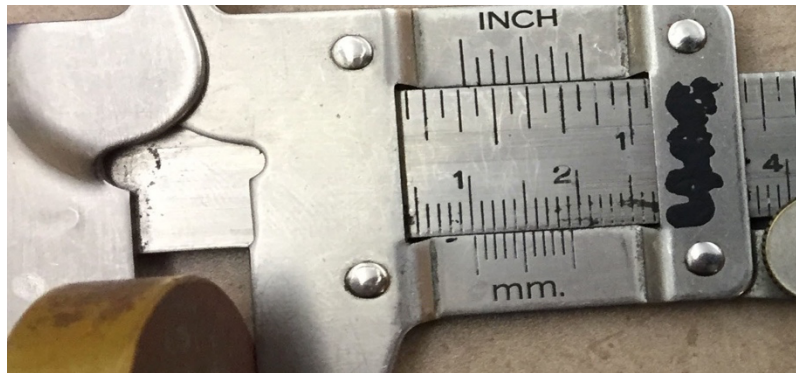


Figure 5

Practice Quiz Answers

Figure 3: 2.590 ± 0.005 cm

Figure 4: 4.140 ± 0.005 cm

Figure 5: 1.020 ± 0.005 cm